Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-6686
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Team Details	**Team Size:** 4 **Team Leader:** MANISHA R **Team Members:** • GOPIKA K • VINDHIYA G • NITHIYASRI M
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, Postman Runner, ServiceNow Automated Test Framework (ATF)
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Service Catalog Order Placement API (`/api/sn_sc/v1/servicecatalog/items/submit`) & Flow Designer Execution Engine
Test Environment	ServiceNow Developer Instance (Cloud Hosted - San Diego / Washington DC Release)
Test Date	12 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Simultaneous Catalog Submissions for Laptop Requests	50	120	Response time < 2.0s without request drops
2	Peak Load Concurrent Approvals by Managers	100	180	Approval flows execute and update DB state within 3.0s
3	Spike Load Test on Inventory Validation API	200	60	System handles sudden traffic without 5xx errors
4	End-to-End Procurement Flow Execution	30	300	All sub-flows complete under 4.0s average

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.18 seconds	**Pass**	Optimal response
2	Response Time (Max)	< 5 seconds	3.42 seconds	**Pass**	Under peak spike
3	Throughput (Req/sec)	> 25 req/sec	38.4 req/sec	**Pass**	Exceeded target
4	Error Rate	< 1%	0.12%	**Pass**	Very low failure
5	CPU Utilization	< 80%	62.5%	**Pass**	Stable server load
6	Memory Utilization	< 80%	54.1%	**Pass**	No memory leaks

Step 4: Observations & Analysis

Key Findings:
The automated laptop procurement workflow built with ServiceNow Flow Designer handled concurrent catalog requests efficiently. Under a normal load of 50 users, response time averaged 1.18 seconds. Under peak load of 100 concurrent approvals, the system maintained an throughput of 38.4 req/sec with an error rate well below 0.2%.

Bottlenecks Identified:
Initial test runs exhibited minor latency during complex multi-tier approval lookup queries when searching unindexed user hierarchy fields in the Sys User table.

Optimization Steps Taken:
Added database index on `sys_user.manager` field, enabled asynchronous flow execution for external notification tasks (Teams/Email), and cached inventory stock levels in ServiceNow MID server memory.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here - JMeter / Postman Performance Summary Graph]

[Insert Screenshot 2 here - ServiceNow Transaction & Response Log Graph]